## Type MS · The final 20-mark program

### Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 1 of 7 · find part whole in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 | Award **one** mark for each of: | | | |
| | A1 | Any variable for tracking best breed | Initial values could be the first breed in the table | |
| | A2 | A loop to all items in an array | The same index can be used across all the individual arrays, regardless of loop type | |
| | A3 | Daily volume calculation is volume * count | Disregard accuracy of indexing | |
| | A4 | Rows of arrays are displayed | Disregard presence/lack of calculated daily volume | |
| | A5 | Total volume calculated correctly by any method | | |
| | A6 | Relational operators used to compare rating and daily volume | | |
| | A7 | Boolean operator/nested selection used to combine tests | | |
| | A8 | Add calculated daily volume to the new data structure | | |
| | A9 | Description of data fields displayed (no alignment required) | | |
| | A10 | Recommended breed identified correctly | by any method except hard-coded | |
| | A11 | Outputted information is fit for purpose and suitable for the audience | | **11** |

## Type MS · The final 20-mark program

### Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 2 of 7 · find part whole in this mark scheme extract

| Award up to a maximum of nine marks using the levels-based mark scheme below. | | | | Mark |
|---|---|---|---|---|
| **Band 0** | **Band 1 (1-3 marks)** | **Band 2 (4-6 marks)** | **Band 3 (7-9 marks)** | |
| No rewardable content | Little attempt to decompose the problem into component parts | Some attempt to decompose the problem into component parts | The problem has been decomposed into component parts | |
| | Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| | Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| | Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| | Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| | The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | **(9)** |

## Type MS · The final 20-mark program

### Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 3 of 7 · find part whole in this mark scheme extract

**C#**

```
// Q06FINISHED

string[] tbl_breed = { "Red Chittagong", "Sussex", "Dexter",
                       "Abondance", "Sahiwal", "Vorderwald",
                       "Ayrshire", "Jersey", "Randall",
                       "Alderney", "Carora", "Gloucester"};
int[] tbl_rating = { 1, 2, 3, 2, 3, 1, 2, 1, 2, 1, 3, 2 };
int[] tbl_count = { 6, 3, 8, 7, 6, 4, 3, 7, 3, 3, 4, 7 };
double[] tbl_volume = {7.5, 5.7, 11.4, 11.4,
                       22.0, 15.2, 21.0, 18.3,
                       19.0, 9.0, 23.1, 16.0 };
double[] tbl_dailyVolume = {0.0, 0.0, 0,0, 0,0,
                            0.0, 0.0, 0.0, 0.0,
                            0.0, 0.0, 0.0, 0.0};

// ---------------------------------------------
// Write your code below this line

const String SPACE = " ";           // Just for reading

// Variables for indexing and totals
double totalVolume = 0.0;
double todayVolume = 0.0;

// Variables for outputting
string outString = "";

// Initialise the maximum values to the first instance
int maxIndex = 0;

// Display a message to describe the data fields
Console.WriteLine ("Fields are: Breed, Rating, Volume (cow), Count, Volume (day)");
```

```
// Process every breed of cow
for (int i = 0; i < tbl_breed.Length; i++)
{
    // Calculate volume per day
    todayVolume = tbl_volume[i] * tbl_count[i];

    // Add today's volume to the daily volume table
    tbl_dailyVolume[i] = todayVolume;

    // Display the row of information for this breed
    outString = tbl_breed[i] + SPACE +
                tbl_rating[i].ToString() + SPACE +
                tbl_volume[i].ToString() + SPACE +
                tbl_count[i].ToString() + SPACE +
                tbl_dailyVolume[i].ToString();
    Console.WriteLine(outString);

    // Keep a running total of milk production
    totalVolume = totalVolume + todayVolume;

    // Test for a best breed
    if ((tbl_rating[i] <= tbl_rating[maxIndex]) &
        (tbl_volume[i] >= tbl_volume[maxIndex]))
    {
        maxIndex = i;
    }
}

// Display the total volume of milk in a day
outString = "Total: " + totalVolume.ToString() + " litres";
Console.WriteLine(outString);

// Display the recommended breed
outString = "Recommended breed: " + tbl_breed[maxIndex] +
            " rating: " + tbl_rating[maxIndex].ToString() +
            " volume: " + tbl_volume[maxIndex];
Console.WriteLine(outString);
```

## Type MS · The final 20-mark program

### Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 4 of 7 · find part whole in this mark scheme extract

**Java**

```java
// Q06FINISHED

public class Q06FINISHED
{
    static final String SPACE = " ";            // Just for reading

    public static void main(String[] args)
    {
        String[] tbl_breed = { "Red Chittagong", "Sussex", "Dexter", "Abondance",
                               "Sahiwal", "Vorderwald", "Ayrshire", "Jersey",
                               "Randall", "Alderney", "Carora", "Gloucester"};
        int[] tbl_rating = { 1, 2, 3, 2, 3, 1, 2, 1, 2, 1, 3, 2 };
        int[] tbl_count = { 6, 3, 8, 7, 6, 4, 3, 7, 3, 3, 4, 7 };
        double[] tbl_volume = { 7.5, 5.7, 11.4, 11.4,
                                22.0, 15.2, 21.0, 18.3,
                                19.0, 9.0, 23.1, 16.0 };
        double [] tbl_dailyVolume = {0.0, 0.0, 0,0, 0,0,
                                     0.0, 0.0, 0.0, 0.0,
                                     0.0, 0.0, 0.0, 0.0};

        // -----------------------------------------------
        // Write your code below this line

        // Variables for indexing and totals
        double totalVolume = 0.0;
        double todayVolume = 0.0;

        // Variables for outputting
        String outString = "";

        // Initialise the maximum values to the first instance
        int maxIndex = 0;

       // Display the titles in a table
        System.out.println ("Fields are: Breed, Rating, Volume (cow), Count, Volume (day)");

```

## Type MS · The final 20-mark program

### Answer 1 of 8 · Question 6 part whole · 20 marks

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```
37          // Process every breed of cow
38          for (int i = 0; i < tbl_breed.length; i++)
39          {
40              // Calculate volume per day
41              todayVolume = tbl_volume[i] * tbl_count[i];
42
43              // Add today's volume to the daily volume table
44              tbl_dailyVolume[i] = todayVolume;
45
46              // Display the row of information for this breed
47              outString = tbl_breed[i] + SPACE +
48                          Integer.toString(tbl_rating[i]) + SPACE +
49                          String.valueOf(tbl_volume[i]) + SPACE +
50                          Integer.toString(tbl_count[i]) + SPACE +
51                          String.valueOf(tbl_dailyVolume[i]);
52              System.out.println (outString);
53
54              // Keep a running total of milk production
55              totalVolume = totalVolume + todayVolume;
56
57              // Test for a best breed
58              if ((tbl_rating[i] <= tbl_rating[maxIndex]) &
59                      (tbl_volume[i] >= tbl_volume[maxIndex]))
60              {
61                  maxIndex = i;
62              }
63          }
64
65          // Display the total volume of milk in a day
66          outString = "Total: " + Double.toString(totalVolume) + " litres";
67          System.out.println (outString);
68
69          // Display the recommended breed
70          outString = "Recommended breed: " + tbl_breed[maxIndex] +
71                      " rating: " +
72                      Integer.toString(tbl_rating[maxIndex]) +
73                      " volume: " +
74                      Double.toString(tbl_volume[maxIndex]);
75          System.out.println (outString);
76      }
77  }
```

## Type MS · The final 20-mark program

### Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 6 of 7 · find part whole in this mark scheme extract

**Python**

```
# Q06FINISHED

tbl_breed = ["Red Chittagong", "Sussex", "Dexter", "Abondance",
             "Sahiwal", "Vorderwald", "Ayrshire", "Jersey",
             "Randall", "Alderney", "Carora", "Gloucester"]
tbl_rating = [1, 2, 3, 2, 3, 1, 2, 1, 2, 1, 3, 2]
tbl_count = [6, 3, 8, 7, 6, 4, 3, 7, 3, 3, 4, 7]
tbl_volume = [7.5, 5.7, 11.4, 11.4,
              22.0, 15.2, 21.0, 18.3,
              19.0, 9.0, 23.1, 16.0]
tbl_dailyVolume = []

# ---------------------------------------------
# Write your code below this line

# Variables for indexing and totals
index = 0
totalVolume = 0.0

# Initialise the maximum values to the first instance
maxIndex = 0

# Display a message to describe the data fields
print ("Fields are: Breed, Rating, Volume (cow), Count, Volume (day)")

# Process every breed of cow
for index in range (len (tbl_breed)):
    # Calculate volume per day
    todayVolume =  tbl_volume[index] * tbl_count[index]

    # Add today's volume to the daily volume table
    tbl_dailyVolume.append (todayVolume)

    # Display the row of information for this breed
    print (tbl_breed[index],tbl_rating[index],
           tbl_volume[index], tbl_count[index], tbl_dailyVolume[index])

    # Keep a running total of milk production
    totalVolume = totalVolume + todayVolume

    # Test for a best breed
    if ((tbl_rating[index] <= tbl_rating[maxIndex]) and
                (tbl_volume[index] >= tbl_volume[maxIndex])):
        maxIndex = index

# Display the total volume of milk in a day
layout = "Total: {} litres"
print (layout.format (totalVolume))

# Display the recommended breed
print ("Recommended breed: " + tbl_breed[maxIndex],
       "rating: " + str(tbl_rating[maxIndex]),
       "volume " + str(tbl_volume[maxIndex]))
```

## Type MS · The final 20-mark program

### Answer 1 of 8 · Question 6 part whole · 20 marks

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## Type MS · The final 20-mark program

### Answer 2 of 8 · Question 5 part whole · 20 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

| | | | |
|---|---|---|---|
| Q5 | Designated marks:<br>1. Initialise variables for at least two of total number of books, count of pupils and average<br>2. Initialise variables for gold, silver and bronze / sort array in order of books read<br>3. Create loop for identifying (and printing) those borrowing fewer than 10 books<br>4. Print out appropriate details (must include pupil ID)<br>5. Print out total of books borrowed<br>6. Print out average number of books borrowed<br>7. Identification of gold medal winner<br>8. Identification of silver medal winner<br>9. Identification of bronze medal winner<br>10. Details of at least one of gold, silver and bronze medal winner printed (minimum last name or first name)<br>11. Details of all three medal winners (first name and last name) | Award marks independently<br><br>Award marks for reasonable attempt to meet requirement. | (11) |

## Type MS · The final 20-mark program

### Answer 2 of 8 · Question 5 part whole · 20 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

| **Question 5 Overview marks:** | | | | |
|---|---|---|---|---|
| **Mark Band 0** | **Mark Band 1** | **Mark Band 2** | **Mark Band 3** | **Marks** |
| 0 | 1 – 3 | 4 – 6 | 7 – 9 | |
| No rewardable content | Little attempt to decompose the problem into component parts<br><br>Some parts of the logic are clear and appropriate to the problem<br><br>Some appropriate use and manipulation of data types, variables, data structures and program constructs<br><br>Parts of the code are clear and readable<br><br>Finished program will not be flexible enough with other data sets or input<br><br>The program meets some of the given requirements | Some attempt to decompose the problem into component parts<br><br>Most parts of the logic are clear and mostly appropriate to the problem<br><br>The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate<br><br>Code is mostly clear and readable<br><br>Finished program will function with some but not all other data sets or input<br><br>The program meets most of the given requirements | The problem has been decomposed into component parts<br><br>The logic is clear and appropriate to the problem<br><br>The use and manipulation of data types, variables and data structures and program constructs is appropriate<br><br>Code is clear and readable<br><br>Finished program could be used with other data sets or input<br><br>The program fully meets the given requirements | (9) |

## Type MS · The final 20-mark program

### Answer 2 of 8 · Question 5 part whole · 20 marks

2019 JUN · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

**C# Example code**

## Type MS · The final 20-mark program

### Answer 3 of 8 · Question 6 part whole · 20 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 6 | A1 | 2.2.1 | Initialise variable for total sales by member of staff (1) | Sales for each member of staff<br><br>George Taylor 28466<br>Fehlix Chayne 35250<br>Sumren Bergen 30186<br>Samira Beckle 37430<br>Nellie Bogart 20448<br>Cheryl Grouth 34383<br>Danuta Graunt 26421<br>Jaiden Deckle 21507<br>Jimran Caliks 32415<br>Deynar Derran 28590<br>==========<br><br>Total staff sales: 295096<br><br>Highest sales by: Samira Beckle with 37430<br><br>Second highest sales by: Fehlix Chayne with 35250<br>>>> | (11) |
| | A2 | 2.5.1 | Add succeeding months' sales to total sales (1) | | |
| | A3 | 2.1.6 | Display total sales by member of staff (1) | | |
| | A4 | 2.2.2 | Loop repeats for each member of staff (1) | | |
| | A5 | 2.2.1 | Running team total sales initialised (1) | | |
| | A6 | 2.2.2 | Add succeeding individual staff sales to team total (1) | | |
| | A7 | 2.1.6 | Display final running total (1) | | |
| | A8 | 2.1.6 | Staff member with highest individual sales identified (1) | | |
| | A9 | 2.1.6 | Staff member with second highest individual sales identified (1) | | |
| | A10 | 2.1.6 | Information about staff with highest sales displayed (1) | | |
| | A11 | 2.1.6 | Information about staff with second highest sales displayed (1) | | |

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## Type MS · The final 20-mark program

### Answer 3 of 8 · Question 6 part whole · 20 marks

2020 NOV · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

| Band 1 (1-3 marks) | Band 2 (4-6 marks) | Band 3 (7-9 marks) | Mark |
|---|---|---|---|
| Little attempt to decompose the problem into component parts | Some attempt to decompose the problem into component parts | The problem has been decomposed into component parts | |
| Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | (9) |

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## Type MS · The final 20-mark program

### Answer 3 of 8 · Question 6 part whole · 20 marks

2020 NOV · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Code example

Python

```
allStaffTotal = 0
highestSales = 0
secondSales = 0
high = 0
second = 0

print ("Sales for each member of staff","\n\n")
for staff in staffSales:
    staffTotal = 0
    sales = 3
    while (sales < 9):
        staffTotal = staffTotal + staff[sales]
        sales = sales + 1
    print (staff[1],staff[2],staffTotal)
    if (staffTotal > secondSales):
        if (staffTotal > highestSales):
            highestSales = staffTotal
            high = staff
        else:
            secondSales = staffTotal
            second = staff
    allStaffTotal = allStaffTotal + staffTotal
print ("==========\n\nTotal sales for all staff: ", allStaffTotal)

print ("\n\nHighest sales by: ",high[1],high[2], " with ", highestSales)
print ("\n\nSecond highest sales by: ",second[1],second[2], " with ", secondSales)
```

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## Type MS · The final 20-mark program

### Answer 4 of 8 · Question 5 part whole · 20 marks

2021 JUN · Paper 02 · extract 1 of 5 · find part whole in this mark scheme extract

| Question | | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 5 | | **addPlayerName()** | | |
| | A1 | Suitable prompt for player name and assigned to suitable variable | | |
| | | **guessCapital()** | | |
| | A2 | Ensure question can only be used once | Do not award if more than one question variable e.g. question1, question2 etc. | |
| | A3 | Question includes suitable message and country name | | |
| | A4 | Check made to see if guess is correct | | |
| | A5 | If guess correct score incremented | | |
| | A6 | If guess is incorrect suitable message displayed | | |
| | A7 | If guess incorrect capital concatenated with message | | |
| | A8 | Repeated for a minimum of five questions | Do not award if questions are asked manually e.g. question1, question2, repeated code for each question etc.<br><br>Do not award if 5 unique questions are not asked while the program is running | |
| | | **Main Program** | | |
| | A9 | Player name or score displayed | Do not award if the return value from at least one function is not used | |
| | A10 | At least one menuChoice calls correct subprogram | | |
| | A11 | Main program calls the two sub-programs correctly | | (11) |

## Type MS · The final 20-mark program

### Answer 4 of 8 · Question 5 part whole · 20 marks

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| Band 1 (1-3 marks) | Band 2 (4-6 marks) | Band 3 (7-9 marks) | Mark |
|---|---|---|---|
| Little attempt to decompose into component parts | Some attempt to decompose into component parts | The problem has been decomposed into component parts | |
| Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | (9) |

## Type MS · The final 20-mark program

### Answer 4 of 8 · Question 5 part whole · 20 marks

2021 JUN · Paper 02 · extract 3 of 5 · find part whole in this mark scheme extract

Code examples

Python

**Add player name function**

```
    player = ""
    while(player == ""):
        player = input("Enter your player name ")
    return player
#-----------------------------------------------------
```

**Main program**

```
menuChoice = 0
score = 0
playerName = ""

while menuChoice != 3:
    displayMenu()
    menuChoice = getMenuChoice()

    # Add your code to:
    #   call the relevant subprogram if the menu choice is 1 or 2
    #   display the player name and the score if the menu choice is 3
    if menuChoice == 1:
        playerName = addPlayerName()
    elif menuChoice == 2:
        score = guessCapital()
    else:
        print("Well done "+ playerName + ". The score is " + str(score))
```

## Type MS · The final 20-mark program

### Answer 4 of 8 · Question 5 part whole · 20 marks

2021 JUN · Paper 02 · extract 4 of 5 · find part whole in this mark scheme extract

**Guess capital city function**

```
# Add your code here
questionCount = 1
questionScore = 0

# Ask 5 questions
while questionCount <= 5:

    questionChoice = -1
    questionNumbers = ""

    # Build a string containing the question numbers available
    for question in questions:
        if question != 0:
         questionNumbers += str(question) + " "

    # Ensure valid question number is chosen
    while str(questionChoice) not in questionNumbers:
        questionChoice = int(input("Pick a number from " + questionNumbers))

    # Get the country and its capital
    country = countries[questionChoice - 1]
    capital = capitals[questionChoice - 1]

    # Display the country and get the guess
    guess = input("What is the capital of " + country + "? ").lower()

    # If the guess is correct display message and increment score
    if guess == capital.lower():
        print("Well done, you guessed correctly")
        questionScore += 1
    else:
        # Otherwise display the country name and correct capital
        print("You did not guess correctly. The capital of " + country + " is " + capital)

    # Increment the number of questions asked
    questionCount = questionCount + 1

    # Set the question number to 0 so that it cannot be guessed again
    questions[questionChoice - 1] = 0;
    questions[questionChoice - 1] = 0;

# return the score to the main menu
return questionScore
```

## Type MS · The final 20-mark program

### Answer 4 of 8 · Question 5 part whole · 20 marks

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## Type MS · The final 20-mark program

### Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 1 of 8 · find part whole in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 5 | A1 | At least three variables initialised appropriately | | (1) |
| | A2 | Word input | | (1) |
| | A3 | Program repeats until 1 is input | | (1) |
| | A4 | Loop through each word in the array | | (1) |
| | | Words that begin with the same letter | | |
| | A5 | Check to see if the first letter of the input word matches the first letter of at least one word in the array | | (1) |
| | A6 | The number of words that begin with the same letter displayed | | (1) |
| | | Words that contain the input word | | |
| | A7 | Check to see if the word contains the input word | | (1) |
| | A8 | Number of letters in the word calculated | | (1) |
| | A9 | At least one word from the array that contains the input word displayed | | (1) |
| | A10 | The number of words that contain the input word displayed | | (1) |
| | A11 | Identification of the longest word or the shortest word | | (1) |

| Band 1 (1-3 marks) | Band 2 (4-6 marks) | Band 3 (7-9 marks) | Mark |
|---|---|---|---|
| Little attempt to decompose into component parts | Some attempt to decompose into component parts | The problem has been decomposed into component parts | |
| Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
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| Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | (9) |

## Type MS · The final 20-mark program

### Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 2 of 8 · find part whole in this mark scheme extract

Code examples

C#

```
// Add your code here
string inputWord = "";

while (inputWord != "1")
{
    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    Console.WriteLine("Enter a word or 1 to exit ");
    inputWord = Console.ReadLine();

    if (inputWord != "1")
    {
        int count = 0;

        foreach (String word in wordArray)
        {
            if(word[0] == inputWord[0])
            {
                Console.WriteLine(word);
                count ++;
            }
        } // End of Loop

        Console.WriteLine(count + " word(s) beginning with "+ inputWord[0]);
        Console.WriteLine("----------------------------------------------------------------------");

        count = 0;

        foreach (String word in wordArray)
        {
            if(word.Contains(inputWord))
            {
                count ++;
                Console.WriteLine(word);

                int length = word.Length;

                if (longest < length)
```

## Type MS · The final 20-mark program

### Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 3 of 8 · find part whole in this mark scheme extract

```
                    {
                        longest = length;
                        longestWord = word;
                    }

                    if (shortest > length)
                    {
                        shortest = length;
                        shortestWord = word;
                    }
                }// end of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word

            if (count > 0)
            {
                Console.WriteLine(count + " word(s) with "+ inputWord + " in them");
                Console.WriteLine("The longest word has " + longest + " characters");
                Console.WriteLine("The shortest word has " + shortest + " characters");
                Console.WriteLine("The longest word is " + longestWord + " characters");
                Console.WriteLine("The shortest word is " + shortestWord + " characters");
                Console.WriteLine("--------------------------------------------------------------------------------");
            }
            else
            {
                Console.WriteLine("There were 0 words that had all the letters from " + inputWord + " in them");
                Console.WriteLine("--------------------------------------------------------------------------------");
            } // End of printing words
        } // End of check
    } // End of while loop
```

## Type MS · The final 20-mark program

### Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 4 of 8 · find part whole in this mark scheme extract

Java

```
// Add your code here
Scanner input = new Scanner(System.in);
String inputWord = "";

// While there is a word run the program
while (!inputWord.equals("1")) {

    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    System.out.println("Enter a word or 1 to exit ");
    System.out.println("--------------------------------------------------------------------------------");
    inputWord = input.nextLine();
    if (!inputWord.equals("1")){
        // Find words that begin with the same letter as the input word
        int count = 0;

        // Get each word in the array
        for (String word: wordArray){

            // If the first character of the word is the same as the first character of
            // the input word then print the word
            if (word.charAt(0) == inputWord.charAt(0)){
                System.out.println(word);
                count ++;
            }
        } // End of for loop

        // Print the total number of words that begin with the same letter
        System.out.println(count + " word(s) beginning with "+ inputWord.charAt(0));
        System.out.println("--------------------------------------------------------------------------------");

        // Find words that contain the input word
        count = 0;

        // Get each word in the array
        for (String word: wordArray){
            // Check to see if the word contains the inputWord
            if (word.contains(inputWord)){
                count ++;
                System.out.println(word);

                // Set the length of the word
                int length = word.length();
```

## Type MS · The final 20-mark program

### Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 5 of 8 · find part whole in this mark scheme extract



```
                    // If the length is longer than the current longest replace it and the longestWord
                    if (longest < length) {
                        longest = length;
                        longestWord = word;
                    }

                    // If the length is shorter than the shortest replace it and the shortestWord
                    if (shortest > length){
                        shortest = length;
                        shortestWord = word;
                    }
                } // End of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word
            if( count > 0){
                System.out.println(count + " word(s) with "+ inputWord + " in them");
                System.out.println("The longest word has " + longest + " characters");
                System.out.println("The shortest word has " + shortest + " characters");
                System.out.println("The longest word is " + longestWord + " characters");
                System.out.println("The shortest word is " + shortestWord + " characters");
                System.out.println("--------------------------------------------------------------------------------");
            } else {
                System.out.println("There were 0 words that had all the letters from " + inputWord + " in them");
                System.out.println("--------------------------------------------------------------------------------");
            } // End of printing words
        }
    } // End of while loop
```

## Type MS · The final 20-mark program

### Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 6 of 8 · find part whole in this mark scheme extract

Python

```
# While there is a word run the program
while inputWord !="1":

    shortest = 5000
    longest = 0
    shortestWord = ""
    longestWord = ""

    inputWord = input("Enter a word or 1 to exit: ")
    print("------------------------------------------------------------------------")

    if inputWord !="1":
        # Find words that begin with the same letter as the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # If the first character of the word is the same as the first character of
            # the input word then print the word
            if word[0] == inputWord[0]:
                print(word)
                count = count + 1

        # Print the total number of words that begin with the same letter
        print("\n",count,"word(s) beginning with",inputWord[0])
        print("------------------------------------------------------------------------")

        # Find words that contain the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # Check to see if the word contains the inputWord
            if inputWord in word:
                count = count + 1
                print(word)
                # Set the length of the word
                length = len(word)

                # If the length is longer than the current longest replace it and the longestWord
                if longest < length:
                    longest = length
                    longestWord = word

                # If the length is shorter than the shortest replace it and the shortestWord
                if shortest > length:
                    shortest = length
                    shortestWord = word
```

## Type MS · The final 20-mark program

### Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 7 of 8 · find part whole in this mark scheme extract

```
59
60          # Print the number of words, the number of characters in the longest and
61          # shortest word, the longest and the shortest word
62          if count > 0:
63              print("\n",count,"word(s) with",inputWord,"in them")
64              print("The longest word has",longest,"letters")
65              print("The shortest word has",shortest,"letters")
66              print("The longest word is",longestWord)
67              print("The shortest word is",shortestWord)
68              print("-------------------------------------------------------------------------")
69          else:
70              print("There were 0 words with all the letters from",inputWord,"in them")
71          # ------------------------------------------------------------------------------------
72
73  # End the program when there are no more words
74  print("End of program")
```

## Type MS · The final 20-mark program

### Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 8 of 8 · find part whole in this mark scheme extract

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## Type MS · The final 20-mark program

### Answer 6 of 8 · Question 6 part whole · 20 marks

2022 JUN · Paper 02 · extract 1 of 4 · find part whole in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 | A1 | At least two appropriate subprograms used | | (1) |
| | A2 | Repeat guess until the word has been guessed **OR** no attempts left | | (1) |
| | A3 | Generate and display the number of attempts left | Must see working correctly | (1) |
| | A4 | Request input of the word | | (1) |
| | A5 | Validate the input of the word to ensure it is the same length as the random word | | (1) |
| | A6 | Check to see if the guess matches the random word | | (1) |
| | A7 | If the word has been guessed, display message including the random word and number of attempts | | (1) |
| | A8 | Keep track and display letters that are in the word | Must see working | (1) |
| | A9 | Keep track and display letters that are not in the word | Must see working | (1) |
| | A10 | Letters appear only once in each list | Must see working | (1) |
| | A11 | Display lose message including the random word | | (1) |

| Band 1 (1-3 marks) | Band 2 (4-6 marks) | Band 3 (7-9 marks) | Mark |
|---|---|---|---|
| Little attempt to decompose into component parts | Some attempt to decompose into component parts | The problem has been decomposed into component parts | |
| Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | (9) |

## Type MS · The final 20-mark program

### Answer 6 of 8 · Question 6 part whole · 20 marks

2022 JUN · Paper 02 · extract 2 of 4 · find part whole in this mark scheme extract

Code examples

Python

Subprograms

```
# Add your subprograms here
def checkInput(pWordLength):
    # Validate input to make sure it is the same length as the secret word
    check = ""
    while len(check) != pWordLength:
        check = input("Enter your guess. The secret word has " +str(pWordLength)+ " letters: ")
    return(check)

def checkLettersInWord(pGuess, pWordToGuess, pLetters, type):
    for letter in pGuess: # Check each letter in the guess
        if type == 0: # The check to carry out is for correct letters
            # If the letter is in the secret word and not already in the correctletters, add it
            if letter in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
        else: # The check to carry out is for wrong letters
            # If the letter is not in the secret word and not already in the wrongLetters, add
            if letter not in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
    return pLetters

def display(pCorrectLetters, pWrongLetters):
    # Display the correct and wrong letters
    if len(pCorrectLetters) > 0:
        print("Letter(s) in the secret word:",pCorrectLetters)

    if len(pWrongLetters) > 0:
        print("Letter(s) not in the secret word:",pWrongLetters)
```

## Type MS · The final 20-mark program

### Answer 6 of 8 · Question 6 part whole · 20 marks

2022 JUN · Paper 02 · extract 3 of 4 · find part whole in this mark scheme extract

**Main program**

```
# Add your main program code here
wordLength = len(wordToGuess)
maxAttempts = wordLength + 3
numAttempts = 0
correctLetters = ""
wrongLetters = ""
guessed = False

# Loop until the no attempts are left or the secret word has been guessed
while numAttempts < maxAttempts and guessed == False:
    print("You have",maxAttempts - numAttempts,"attempts to guess the secret word.")
    numAttempts = numAttempts + 1

    guess = checkInput(wordLength)  # Validate the input

    if guess == wordToGuess: # Check to see if the secret word has been guessed
        guessed = True
        print("Well done. You guessed the secret word was",wordToGuess,"in",numAttempts,"attempts.")

    else: # If it hasn't been guessed check for correct and wrong letters
        correctLetters = checkLettersInWord(guess,wordToGuess, correctLetters, 0)
        wrongLetters = checkLettersInWord(guess,wordToGuess, wrongLetters, 1)
        display(correctLetters, wrongLetters)

# Display game over message if attempts have run out and the secret word has not been guessed
if numAttempts == maxAttempts and guessed == False:
    print("Game over. You did not guess that the secret word was",wordToGuess)
```

## Type MS · The final 20-mark program

### Answer 6 of 8 · Question 6 part whole · 20 marks

2022 JUN · Paper 02 · extract 4 of 4 · find part whole in this mark scheme extract

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## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 1 of 10 · find part whole in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 | | Award **one** mark for each of: | | |
| | A1 | Put new group of words (newspaper, magazine) into last blank position in array (1) | Any method<br>Do not award hard-coded changes to table<br>Note: *if* the solution will not work for any length of array. This will impact LBMS functionality. | |
| | A2 | Loop for processing every group of words in the array (1) | Allow hard-coded length | |
| | A3 | Loop controlled by the length of the array (1) | Method/function to determine the number of groups in the array | |
| | A4 | Mechanism for tracking group number to appear in output (1) | Any method | |
| | A5 | Output includes row number and original group of words (1) | Ignore 'punctuation'<br>Requires both a line number and each word pair | |
| | A6 | Mechanism for finding longest word in each group (1) | Method/function to determine the length of each string | |
| | A7 | Mechanism for determining if the words in the group are out of order (1) | Any method | |
| | A8 | Mechanism for putting the words in the group into alphabetical order (1) | Any method | |
| | A9 | Output includes longest word in each group and any group found to be out of order is displayed in order (1) | | |
| | A10 | Evidence of two-dimensional indexing (1) | Any method | **11** |

## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 2 of 10 · find part whole in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | A11 | At least one instance of an informative comment (1) | | |

## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 3 of 10 · find part whole in this mark scheme extract

| Award up to a maximum of nine marks using the levels-based mark scheme below. | | | | Mark |
|---|---|---|---|---|
| **Band 0** | **Band 1 (1-3 marks)** | **Band 2 (4-6 marks)** | **Band 3 (7-9 marks)** | |
| No rewardable content | Little attempt to decompose the problem into component parts | Some attempt to decompose the problem into component parts | The problem has been decomposed into component parts | |
| | Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| | Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| | Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| | Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| | The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | **(9)** |

## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 4 of 10 · find part whole in this mark scheme extract

**C#**

```
// Q06FINISHED

String[,] tblWords = {{"apple", "banana"},
                      {"wrist", "leg"},
                      {"blue", "yellow"},
                      {"speaker", "keyboard"},
                      {"lavender", "tulip"},
                      {"pencil", "chalk"},
                      {"apartment", "house"},
                      {"bottom", "top"},
                      {"snow", "fog"},
                      {"beach", "mountain"},
                      {"", ""}};

String word1 = "newspaper";
String word2 = "book";

// -----------------------------------------------
// Write your code below this line
int numRows = tblWords.GetLength(0);
String outString = "";              // For output
int indexMaxLength = 0;             // For checking lengths
String temp = "";                    // For swapping

// Insert a new group of words into the blank row
tblWords[numRows - 1, 0] = word1;
tblWords[numRows - 1, 1] = word2;

```

## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 5 of 10 · find part whole in this mark scheme extract

```
// Loop for each individual group of words in the table
for (int i = 0; i < numRows; i++)
{
    // Pick up an entire row
    String[] words;
    words = new String[2];
    words[0] = tblWords[i, 0];
    words[1] = tblWords[i, 1];

    // Display the row number and the original content
    outString = Convert.ToString (1 + i);
    outString = outString + " " + words[0] + " " + words[1];
    Console.WriteLine (outString);

    // Find the length of the longest word and display the word
    indexMaxLength = 0;             // Longest is first word
    if (words[1].Length > words[indexMaxLength].Length)
    {
        indexMaxLength = 1;
    }
    // Display indented
    Console.WriteLine ("\t" + words[indexMaxLength]);

    // Check if the words in this group are in order
    if (words[0][0] > words[1][0])
    {
        // This group is out of order so needs ordering
        temp = words[0];
        words[0] = words[1];
        words[1] = temp;

        Console.WriteLine ("\t" + words[0] + " " + words[1]);
    }
}
```

## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 6 of 10 · find part whole in this mark scheme extract

**Java**

```java
// Q06FINISHED

public class Q06FINISHED
{
    public static void main(String[] args)
    {
        String[][] tblWords = {{"apple", "banana"},
                               {"wrist", "leg"},
                               {"blue", "yellow"},
                               {"speaker", "keyboard"},
                               {"lavender", "tulip"},
                               {"pencil", "chalk"},
                               {"apartment", "house"},
                               {"bottom", "top"},
                               {"snow", "fog"},
                               {"beach", "mountain"},
                               {"", ""}};

        String word1 = "newspaper";
        String word2 = "book";

        // --------------------------------------------
        // Write your code below this line
        int numRows = tblWords.length;
        String outString = "";              // For output
        int indexMaxLength = 0;             // For checking lengths
        String temp = "";                    // For swapping

        // Insert a new group of words into the blank row
        tblWords[numRows - 1][0] = word1;
        tblWords[numRows - 1][1] = word2;

```

## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 7 of 10 · find part whole in this mark scheme extract

```
        // Loop for each individual group of words in the table
        for (int i = 0; i < numRows; i++)
        {
            // Pick up an entire row
            String[] words = tblWords[i];

            // Display the row number and the original content
            outString = String.valueOf(1 + i) + " " +  words[0] + " " + words[1];
            System.out.println (outString);

            // Find the length of the longest word and display the word
            indexMaxLength = 0;             // Longest is first word
            if (words[1].length() > words[indexMaxLength].length())
            {
                indexMaxLength = 1;
            }

            // Display indented
            System.out.println ("\t" + words[indexMaxLength]);

            // Check if the words in this group are in order
            if (words[0].charAt (0) > words[1].charAt(0))
            {
                // This group is out of order so needs ordering
                temp = words[0];
                words[0] = words[1];
                words[1] = temp;

                System.out.println ("\t" + words[0] + " " + words[1]);

            }
        }
    }
}
```

## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 8 of 10 · find part whole in this mark scheme extract

**Python**

```
# Q06FINISHED

tblWords = [["apple", "banana"],
            ["wrist", "leg"],
            ["blue", "yellow"],
            ["speaker", "keyboard"],
            ["lavender", "tulip"],
            ["pencil", "chalk"],
            ["apartment", "house"],
            ["bottom", "top"],
            ["snow", "fog"],
            ["beach", "mountain"],
            ["", ""]]

word1 = "newspaper"
word2 = "book"

# ------------------------------------------------
# Write your code below this line
row = 1                 # Row number
numRows = len (tblWords)

# Insert a new group of words into the blank row
tblWords[numRows - 1][0] = word1
tblWords[numRows - 1][1] = word2

# Loop for each individual group of words in the table
for words in tblWords:
    # Display the row number and the original content
    print (str (row) + " " + words[0] + " " + words[1])

    # Find the length of the longest word and display the word
    indexMaxLength = 0          # Longest is first word
    if (len (words[1]) > len (words[indexMaxLength])):
        indexMaxLength = 1

    # Display indented
    print (" "*5 + words[indexMaxLength])

    # Check if the words in this group are in order
    if (words[0] > words[1]):
        # This group is out of order so needs sorting
        temp = words[0]
        words[0] = words[1]
        words[1] = temp
```

## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 9 of 10 · find part whole in this mark scheme extract

**Output**

```
1 apple banana
    banana
2 wrist leg
    wrist
    leg wrist
3 blue yellow
    yellow
4 speaker keyboard
    keyboard
    keyboard speaker
5 lavender tulip
    lavender
6 pencil chalk
    pencil
    chalk pencil
7 apartment house
    apartment
8 bottom top
    bottom
9 snow fog
    snow
    fog snow
10 beach mountain
```

## Type MS · The final 20-mark program

### Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 10 of 10 · find part whole in this mark scheme extract

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## Type MS · The final 20-mark program

### Answer 8 of 8 · Question 6 part whole · 20 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part whole in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 | | Award **one** mark for each of: | | |
| | A1 | Both City and Temp taken as input (1) | | |
| | A2 | Both correct conversion calculation coded (1) | Does not have to be used | |
| | A3 | Any relevant subprogram has been declared (1) | At least one **relevant** subprogram declared.<br>`def name(param):` or<br>`def name():`<br>**and** one line of code indented | |
| | A4 | Temperature in correct parallel position is updated when existing city is input (1) | | |
| | A5 | Loop for accessing all items in either array (1) | **For each** acceptable for this mark | |
| | A6 | Table includes header and dividing line (1) | `print("city Celsius fahrenheit")`<br>`print("-------------")` | |
| | A7 | Mechanism for getting temp from string and completing conversion (1) | Include int( ) and passed to formula for at least one temperature value | |
| | A8 | Mechanism for getting scale C/F (1) | | |
| | A9 | All rows output to table with corresponding city and (at least one) temperature values (1) | (same index value used)<br>do not accept hard coded | |
| | A10 | Table (for cities) all correct (1) | Ignore header (A6) and average (A11)<br><br>Must have city, C value and F value (all accurate)<br><br>Ignore float values if calculation correct | |
| | A11 | Overall average correctly calculated and displayed in both scales (1) | Average may change depending on whether whole number cast or rounded – award accurate formula, not result. | **11** |

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## Type MS · The final 20-mark program

### Answer 8 of 8 · Question 6 part whole · 20 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part whole in this mark scheme extract

Paper 2 – Application of Computational Thinking

LBMS (AOs: 3xAO3.2a, 3xAO3.2b, 3xAO3.2c) (Tgt: 3, 3, 3); Spec: 1.1.6(2), 1.2.1(2), 1.2.2(2), 2.1.1, 2.1.2, 2.1.5
PBMS: 2.1.1, 2.3.2(2), 2.3.3(2), 2.4.1(2), 2.5.1(2), 2.6.1(2)

| Award up to a maximum of nine marks using the levels-based mark scheme below. | | | | Mark |
|---|---|---|---|---|
| **Band 0** | **Band 1 (1-3 marks)** | **Band 2 (4-6 marks)** | **Band 3 (7-9 marks)** | |
| No rewardable content | Little attempt to decompose the problem into component parts | Some attempt to decompose the problem into component parts | The problem has been decomposed into component parts | |
| | Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| | Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| | Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| | Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| | The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | **(9)** |

21 of 23

## Type MS · The final 20-mark program

### Answer 8 of 8 · Question 6 part whole · 20 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part whole in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
# Q06

# city and temperature arrays
Tbl_city = ["Reggane", "Cairo", "New Delhi", "Muscat", "Athens",
            "Barcelona", "Havana", "Phoenix", "Brisbane", "Darwin"]
Tbl_temps = ["F103", "F82", "C34", "F95", "C29", "F79", "F84",
             "C35", "F77", "F81"]

# ----------------------------------------------
# write subprogam(s) below this line
def CtoF(pTemp):
    fhTemp = pTemp * 1.8 + 32
    fhTemp = int(round(fhTemp,0))
    return fhTemp

def FtoC(pTemp):
    ctTemp = (pTemp - 32) / 1.8
    ctTemp = int(round(ctTemp,0))
    return ctTemp

# ----------------------------------------------
# Write your code below this line
totalF = 0
totalC = 0
cTemp = 0
fTemp = 0

cityCount = len(Tbl_city)

# update city temp
thisCity = input("Please enter City: ")
for x in range(cityCount):
    if thisCity == Tbl_city[x]:
        thisTemp = input("Please enter temp in correct format: ")
        Tbl_temps[x] = thisTemp
else:
    print("City does not exist")
```

```
# display table header
print("City          Celsius          Fahrenheit")
print("-----------------------------------------------------")

# take each city and temp
for i in range(cityCount):

    # get each scale and temp
    nextTemp = Tbl_temps[i]
    scale = nextTemp[0]
    theTemp = int(nextTemp[1:])

    # convert between temp scales
    if scale == "C":
        cTemp = theTemp
        totalC = totalC + cTemp
        fTemp = CtoF(theTemp)
        totalF = totalF + fTemp
    else:
        fTemp = theTemp
        totalF = totalF + fTemp
        cTemp = FtoC(theTemp)
        totalC = totalC + cTemp

    # display each row
    print(Tbl_city[i], "          ", cTemp, "          ", fTemp)

# calculate averages
averageF = int(round(totalF / cityCount,0))
averageC = int(round(totalC / cityCount,0))

# display table footer
print("-"*40)
print("Overall Average      ", averageC, "          ", averageF)
```

22 of 23

## Type MS · The final 20-mark program

### Answer 8 of 8 · Question 6 part whole · 20 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part whole in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
# Q06

# city and temperature arrays
Tbl_city = ["Reggane", "Cairo", "New Delhi", "Muscat", "Athens",
            "Barcelona", "Havana", "Phoenix", "Brisbane", "Darwin"]
Tbl_temps = ["F103", "F82", "C34", "F95", "C29", "F79", "F84",
             "C35", "F77", "F81"]

# -----------------------------------------------------
# write subprogam(s) below this line
def CtoF(pTemp):
    fhTemp = pTemp * 1.8 + 32
    fhTemp = int(round(fhTemp,0))
    return fhTemp

def FtoC(pTemp):
    ctTemp = (pTemp - 32) / 1.8
    ctTemp = int(round(ctTemp,0))
    return ctTemp

# -----------------------------------------------------
# Write your code below this line
totalF = 0
totalC = 0
cTemp = 0
fTemp = 0

# update city temp
thisCity = input("Please enter City: ")
if thisCity in Tbl_city:
    thisIndex = Tbl_city.index(thisCity)
    thisTemp = input("Please enter temp in correct format: ")
    Tbl_temps[thisIndex] = thisTemp
else:
    print("City does not exist")
```

```
# display table header
layout = "{:16} {:^10} {:^10}"
print(layout.format("City", "Celsius", "Fahrenheit"))
print("-"*40)

cityCount = len(Tbl_city)

# take each city and temp
for i in range(cityCount):

    # get each scale and temp
    nextTemp = Tbl_temps[i]
    scale = nextTemp[0]
    theTemp = int(nextTemp[1:])

    # convert between temp scales
    if scale == "C":
        cTemp = theTemp
        totalC = totalC + cTemp
        fTemp = CtoF(theTemp)
        totalF = totalF + fTemp
    else:
        fTemp = theTemp
        totalF = totalF + fTemp
        cTemp = FtoC(theTemp)
        totalC = totalC + cTemp

    # display each row
    print(layout.format(Tbl_city[i], cTemp, fTemp))

# calculate averages
averageF = int(round(totalF / cityCount,0))
averageC = int(round(totalC / cityCount,0))

# display table footer
print("-"*40)
print(layout.format("Overall Average", averageC, averageF))
```

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## Type MS · The final 20-mark program

### Answer 8 of 8 · Question 6 part whole · 20 marks

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